fhEVM: A Fully Homomorphic Ethereum Virtual Machine

Lux Partners
research@lux.network

January 2025

Abstract

We present fhEVM, the first production-ready Ethereum Virtual Machine with native support for fully homomorphic encryption. fhEVM enables smart contracts to perform arbitrary computations on encrypted data, achieving confidential DeFi, private governance, and sealed-bid mechanisms without trusted execution environments. We implement fhEVM as a precompile extension to the EVM, achieving 1000× better gas efficiency than naive approaches while maintaining full compatibility with existing Solidity tooling. Our evaluation demonstrates that fhEVM can process 150 encrypted token transfers per second with end-to-end latency under 500ms.

1 Introduction

Public blockchains present a fundamental tension: transparency enables trustless verification but exposes all transaction data to global observers. Financial transactions, governance votes, and auction bids are visible to anyone, enabling front-running, vote buying, and privacy violations.

Current privacy solutions offer limited functionality:

- **Mixers** (Tornado Cash): Hide transaction graphs but cannot compute on encrypted state
- **zkSNARKs**: Prove statements about private inputs but require circuit compilation
- **TEEs**: Rely on trusted hardware with known side-channel vulnerabilities

Key Insight Fully Homomorphic Encryption enables a new paradigm where smart contracts operate directly on ciphertext. Data remains encrypted throughout computation; only authorized parties can decrypt final results.

1.1 Contributions

1. **fhEVM Architecture**: First EVM extension with native encrypted types (euint8, euint16, euint32, euint64, ebool, eaddress)
2. **Gas-Efficient Precompiles**: Dedicated precompiles for FHE operations achieving 1000× improvement over naive EVM implementation
3. **Access Control Layer**: Permissioned decryption with on-chain ACL management
4. **Threshold Key Management**: Distributed decryption eliminating single points of failure
5. **Comprehensive Evaluation**: Benchmarks across DeFi, voting, and auction workloads

2 Background

2.1 Fully Homomorphic Encryption

FHE allows computation on encrypted data without decryption. For a scheme (KeyGen, Enc, Dec, Eval):

Definition 2.1 (Homomorphic Property)**.** For any function f and plaintexts $m_1, \ldots, m_k$:

$$\mathsf{Dec}(\mathsf{sk}, \mathsf{Eval}(\mathsf{evk}, f, \mathsf{Enc}(\mathsf{pk}, m_1), \ldots, \mathsf{Enc}(\mathsf{pk}, m_k))) = f(m_1, \ldots, m_k)$$

We use the TFHE scheme [2] based on the Learning With Errors (LWE) problem:

Definition 2.2 (LWE Problem)**.** Given $(\mathbf{A}, \mathbf{b} = \mathbf{As} + \mathbf{e}) \in \mathbb{Z}_q^{m \times n} \times \mathbb{Z}_q^m$ where $\mathbf{s} \leftarrow \mathcal{U}(\mathbb{Z}_q^n)$ and $\mathbf{e} \leftarrow \chi$ for error distribution χ, distinguish from uniform.

TFHE encrypts bits as elements of the torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$:

$$\mathsf{TLWE}_{\mathbf{s}}(\mu) = (\mathbf{a}, b = \langle \mathbf{a}, \mathbf{s} \rangle + \mu + e) \in \mathbb{T}^{n+1} \quad (1)$$

2.2 Ethereum Virtual Machine

The EVM is a stack-based virtual machine with:

- 256-bit word size
- Persistent storage (key-value mapping)
- Gas metering for computation costs
- Precompiles for expensive operations (e.g., elliptic curve operations)

3 fhEVM Architecture

3.1 Encrypted Types

We introduce native encrypted types to the EVM type system:

Table 1: fhEVM Encrypted Types

Type	Plaintext Range	Ciphertext Size
euint8	$[0, 2^8 - 1]$	8,192 bytes
euint16	$[0, 2^{16} - 1]$	16,384 bytes
euint32	$[0, 2^{32} - 1]$	32,768 bytes
euint64	$[0, 2^{64} - 1]$	65,536 bytes
ebool	$\{0, 1\}$	8,192 bytes
eaddress	$[0, 2^{160} - 1]$	$5 \times$ 32,768 bytes

3.2 FHE Precompiles

We implement FHE operations as EVM precompiles for efficiency:

Table 2: fhEVM Precompile Gas Costs

Address	Operation	Gas Cost
0x0100	FHE.add	65,000
0x0101	FHE.sub	65,000
0x0102	FHE.mul	150,000
0x0103	FHE.lt, FHE.gt	70,000
0x0104	FHE.eq	50,000
0x0105	FHE.select	80,000
0x0106	FHE.rand	100,000
0x0107	Bootstrap	200,000

3.3 Arithmetic Operations

For encrypted integers $\mathsf{ct}_a = \mathsf{Enc}(\mathsf{pk}, a)$ and $\mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, b)$:

$$\mathsf{FHE.add}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxplus \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a + b) \quad (2)$$

$$\mathsf{FHE.sub}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxminus \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a - b) \quad (3)$$

$$\mathsf{FHE.mul}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{ct}_a \boxtimes \mathsf{ct}_b = \mathsf{Enc}(\mathsf{pk}, a \cdot b) \quad (4)$$

Multiplication requires bootstrapping to manage noise:

Theorem 3.1 (Noise Growth)**.** *After L sequential multiplications without bootstrapping:*

$$\mathsf{noise}(\mathsf{ct}) \leq B \cdot (n+1)^L$$

where B is the initial noise bound and n is the LWE dimension.

3.4 Comparison Operations

Comparison produces encrypted booleans:

$$\mathsf{FHE.lt}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Enc}(\mathsf{pk}, \mathbf{1}[a < b]) \quad (5)$$

$$\mathsf{FHE.eq}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Enc}(\mathsf{pk}, \mathbf{1}[a = b]) \quad (6)$$

We implement comparison via the sign function using bootstrapping:

$$\mathsf{FHE.lt}(\mathsf{ct}_a, \mathsf{ct}_b) = \mathsf{Bootstrap}(\mathsf{ct}_a \boxminus \mathsf{ct}_b, f_{\text{sign}}) \quad (7)$$

where $f_{\text{sign}}(x) = 1$ if $x < 0$, else 0.

3.5 Conditional Selection

The select operation enables encrypted branching:

$$\mathsf{FHE.select}(\mathsf{ct}_{\text{cond}}, \mathsf{ct}_a, \mathsf{ct}_b) = (\mathsf{ct}_{\text{cond}} \boxtimes \mathsf{ct}_a) \boxplus ((\mathbf{1} \boxminus \mathsf{ct}_{\text{cond}}) \boxtimes \mathsf{ct}_b) \tag{8}$$

This computes $\mathsf{Enc}(\mathsf{pk}, \text{cond}?a : b)$ without revealing the condition.

4 Access Control

4.1 Permissioned Decryption

Not all parties should decrypt all ciphertexts. We implement on-chain ACLs:

Protocol 4.1 (Permissioned Decryption)**.** 1. Contract owner grants permission: $\mathsf{allow}(\mathsf{addr}_{\text{user}}, \mathsf{ct})$

2. User requests sealed output: $\mathsf{sealOutput}(\mathsf{ct}, \mathsf{pk}_{\text{user}})$
3. Network re-encrypts under user's key
4. User decrypts locally with $\mathsf{sk}_{\text{user}}$

4.2 Formal Security

Theorem 4.2 (Access Control Security)**.** *Under the IND-CPA security of TFHE, an adversary $\mathcal{A}$ without permission cannot distinguish $\mathsf{ct}_0 = \mathsf{Enc}(\mathsf{pk}, m_0)$ from $\mathsf{ct}_1 = \mathsf{Enc}(\mathsf{pk}, m_1)$ with advantage greater than $\mathsf{negl}(\lambda)$.*

5 Threshold Key Management

Single-key FHE creates a central point of failure. We distribute the secret key among n trustees using (t, n) threshold secret sharing based on the RLWE multiparty framework [**?**, **?**].

5.1 Key Generation Ceremony

Protocol 5.1 (Distributed Key Generation)**.** 1. Each trustee T_i generates share $\mathsf{sk}_i \leftarrow \mathcal{N}(0, \sigma^2)^n$

2. Trustees jointly compute $\mathsf{pk} = \sum_{i=1}^{n} \mathsf{pk}_i$
3. Secret key $\mathsf{sk} = \sum_{i=1}^{n} \mathsf{sk}_i$ is never reconstructed

5.2 Threshold Decryption

To decrypt ciphertext $\mathsf{ct} = (\mathbf{a}, b)$:

1. Each trustee computes partial decryption: $d_i = \langle \mathbf{a}, \mathsf{sk}_i \rangle$
2. Combiner aggregates: $d = \sum_{i \in S} \lambda_i \cdot d_i$ for qualified set S
3. Result: $m = \lfloor b - d \rceil$

where λ_i are Lagrange coefficients for the threshold scheme.

6 GPU Acceleration

FHE operations are computationally intensive. We achieve production-grade performance through GPU acceleration [**?**, **?**].

6.1 Metal/CUDA Backend

Our C++ FHE library provides hardware acceleration:

Table 3: GPU-Accelerated FHE Performance

Operation	CPU (ms)	GPU (ms)	Speedup
NTT Forward (N=4096)	0.8	0.04	20×
Bootstrap	50	5	10×
Batch Threshold Decrypt	8,500	302	28×

6.2 Kernel Fusion

Traditional NTT requires $\log_2(N)$ kernel launches. Our fused kernel performs all stages in shared memory:

1. Load coefficients to shared memory (single read)
2. Execute all $\log_2(N)$ butterfly stages with barriers
3. Write results (single write)

This achieves 17× speedup for batch operations by eliminating global memory round-trips.

6.3 Batched Threshold Decryption

For k ciphertexts with (t, n) threshold:

1. **Merkle transcript**: Single tree instead of $O(k)$ serial hashes (70× faster)
2. **Batched partial decrypt**: Vectorized NTT across all ciphertexts (36× faster)
3. **Random linear combination**: Pippenger MSM for verification (22× faster)

Total improvement: 28× for 1000 ciphertext batches.

7 Quantum-Secure Consensus Integration

fhEVM integrates with the Quasar consensus engine [**?**] for quantum-resistant finality.

7.1 Dual-Certificate Finality

Each fhEVM block receives two cryptographic certificates:

1. **BLS Certificate**: Classical security (128-bit)
2. **Ringtail Certificate**: Post-quantum security (Module-LWE, 128-bit PQ) [**?**]

Definition 7.1 (Dual-Certificate Security)**.** A block B is finalized when both certificates are valid:

$$\mathsf{Final}(B) \iff \mathsf{VerifyBLS}(\sigma_{\mathrm{BLS}}, B) \wedge \mathsf{VerifyRingtail}(\sigma_{\mathrm{RT}}, B)$$

This provides defense-in-depth: an attacker must break *both* classical and post-quantum assumptions.

7.2 Ringtail Threshold Signatures

Ringtail [**?**] enables (t, n) threshold signing based on Module-LWE:

- Two-round signing protocol
- ∼1 KB signature per share
- Sub-25ms signing latency
- 128-bit post-quantum security

The same trustees managing FHE key shares also participate in Ringtail signing, providing unified key management.

8 ZAP Protocol Integration

VM-to-node communication uses the Zero-copy Application Protocol (ZAP) [**?**] for minimal overhead.

8.1 Performance Characteristics

Table 4: ZAP vs gRPC Performance

Metric	gRPC	ZAP
Serialization	38 ms/sec	2.5 ms/sec
Memory allocs	107 MB/sec	0 B/sec
Parse latency (1MB)	103,838 ns	28 ns

8.2 FHE-Specific Benefits

For fhEVM workloads at 1000 blocks/sec:

- **BuildBlock**: 6.6× faster ciphertext serialization
- **ParseBlock**: 789× faster ciphertext deserialization
- **Zero-copy**: Ciphertexts read directly from network buffers

9 Applications

9.1 Confidential ERC20 (FHERC20)

Listing 1: FHERC20 Token Contract

```
contract FHERC20 {
    mapping(address => euint64)
       private _balances;

    function transfer(address to,
       euint64 amount)
        external returns (bool)
    {
        _balances[msg.sender] = FHE.
           sub(
```

```
            _balances[msg.sender],
                amount);
        _balances[to] = FHE.add(
            _balances[to], amount);
        return true;
    }

    function balanceOf(address owner
        , bytes32 key)
        external view returns (bytes
            memory)
    {
        require(hasPermission(owner,
            msg.sender));
        return FHE.sealOutput(
            _balances[owner], key);
    }
}
```

9.2 Sealed-Bid Auction

Listing 2: Encrypted Auction

```
contract BlindAuction {
    euint64 private highestBid;
    eaddress private highestBidder;

    function bid(euint64 amount)
        external {
        ebool isHigher = FHE.gt(
            amount, highestBid);
        highestBid = FHE.select(
            isHigher, amount,
                highestBid);
        highestBidder = FHE.select(
            isHigher,
            FHE.asEaddress(msg.
                sender),
            highestBidder);
    }
}
```

10 Evaluation

10.1 Microbenchmarks

10.2 Application Benchmarks

10.3 Comparison with Alternatives

Table 5: FHE Operation Latency

Operation	Latency (ms)	Gas	TPS
Add/Sub	0.5	65,000	2,000
Multiply	12	150,000	83
Compare	8	70,000	125
Select	4	80,000	250
Bootstrap	45	200,000	22

Table 6: End-to-End Application Performance

Application	TPS	Latency (ms)	Gas/Tx
FHERC20 Transfer	150	180	450,000
Sealed Bid	80	320	750,000
Encrypted Vote	200	150	380,000

11 Security Analysis

11.1 Threat Model

We consider an adversary who:

- Controls up to $t-1$ of n threshold key holders
- Observes all blockchain transactions
- Can execute arbitrary smart contracts

11.2 Security Properties

Theorem 11.1 (Ciphertext Indistinguishability)**.** *Under the LWE assumption with parameters (n, q, χ), for any PPT adversary $\mathcal{A}$:*

$$|\mathsf{Pr}[\mathcal{A}(\mathsf{Enc}(\mathsf{pk}, m_0)) = 1] - \mathsf{Pr}[\mathcal{A}(\mathsf{Enc}(\mathsf{pk}, m_1)) = 1]| \leq \mathsf{negl}(\lambda)$$

Theorem 11.2 (Computation Privacy)**.** *Intermediate values during FHE evaluation reveal nothing about inputs beyond what is computable from the output.*

12 Related Work

Homomorphic Encryption Systems Microsoft SEAL [4] and OpenFHE [1] provide FHE libraries but no blockchain integration. The Lattice library [**?**] provides multiparty homomorphic encryption from RLWE [**?**] with efficient threshold protocols [**?**], forming the cryptographic foundation for fhEVM.

Table 7: Privacy Solution Comparison

System	Privacy	Programmable	Decentralized	TEE-free
Tornado Cash	Partial	✗	✓	✓
zkSync	Partial	Limited	✓	✓
Secret Network	Full	✓	Partial	✗
fhEVM	**Full**	✓	✓	✓

Blockchain Privacy Tornado Cash provides transaction privacy via mixing; zkSNARKs enable succinct proofs but limited programmability; Secret Network uses TEEs with known vulnerabilities.

Academic Foundations Our threshold decryption builds on Mouchet et al.'s work on multiparty homomorphic encryption [**?**] and efficient threshold access structures [**?**]. The underlying FHE scheme uses TFHE [2] with LMKCDEY bootstrapping optimizations [**?**] and techniques from the BFV [3] and BGV [**?**] literature.

GPU Acceleration Prior work on GPU-accelerated FHE [**?**] demonstrated order-of-magnitude speedups for basic operations. Our LuxFHE C++ library [**?**] extends this with fused NTT kernels and batched threshold decryption protocols achieving 28× improvement for MPC workloads.

Post-Quantum Consensus Quasar [**?**] provides the first production quantum-secure consensus with dual-certificate finality. Ringtail [**?**] enables practical lattice-based threshold signatures that share key management infrastructure with threshold FHE.

13 Conclusion

fhEVM demonstrates that practical confidential smart contracts are achievable with fully homomorphic encryption. Our implementation processes 150 encrypted token transfers per second while providing cryptographic privacy guarantees superior to TEE-based approaches. The threshold key management eliminates single points of failure, making fhEVM suitable for production deployment.

Availability Implementation available at https://github.com/luxfi/fhe.

References

[1] Ahmad Al Badawi et al. OpenFHE: Open-source fully homomorphic encryption library, 2022.

[2] Ilaria Chillotti, Nicolas Gama, Mariya Georgieva, and Malika Izabachène. TFHE: Fast fully homomorphic encryption over the torus. In *ASIACRYPT*, 2016.

[3] Junfeng Fan and Frederik Vercauteren. Somewhat practical fully homomorphic encryption. In *IACR Cryptology ePrint Archive*, 2012.

[4] Microsoft Research. Microsoft SEAL. https://github.com/microsoft/SEAL, 2023.